

TURING MACHINE

Standard Turing Machine

Turing Machine can be defined as $(Q, \Sigma, \Gamma, \delta, q_0, B, F)$

$Q \Rightarrow$ finite set of states

$\Sigma \Rightarrow$ finite set of i/p alphabets

$\Gamma \Rightarrow$ finite set of tape alphabets

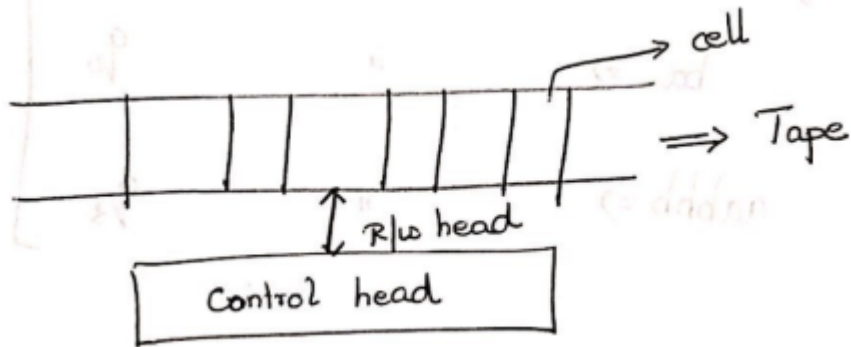
$\delta \Rightarrow \delta: (Q \times \Gamma) \rightarrow (Q \times \Gamma \times \{L, R\})$

$q_0 \in Q \Rightarrow$ Initial state

$B \in \Gamma \Rightarrow$ is used to represent blank

$F \subseteq Q \Rightarrow$ set of final states.

partially
deterministic



Language acceptance:

The language of the TM $(Q, \Sigma, \Gamma, \delta, B, F)$ M
can be defined as...

$$L(M) = \{ W \mid W \text{ in } \Sigma^* \text{ and } q_0 W \xrightarrow{M}^* \alpha_1 p \alpha_2$$

For some p in F , and α_1 and α_2 in Γ^*

ii) Language Acceptability by TM

Language accepted by TM M , denoted as $L(M)$ is the set of strings in Σ^* that cause M to enter a final state or halt state.

Formally language accepted by M is defined as

$$L(M) = \{ w / w \text{ in } \Sigma^* \text{ and } q_0 w \xrightarrow{*} \alpha_1 p \alpha_2 \text{ for some } p \text{ in } F, \\ \text{and } \alpha_1 \text{ and } \alpha_2 \text{ in } \Gamma^* \}$$

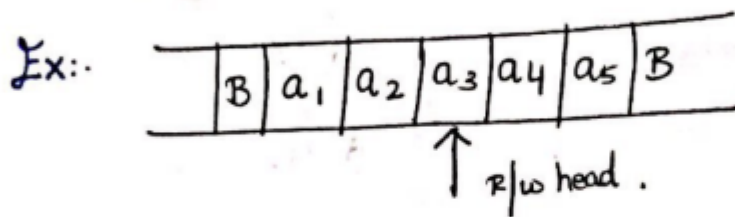
Language that is accepted by TM is said to be recursively enumerable. The term enumerable derives from the fact that it is precisely these languages whose strings can be enumerated by a TM. The class of recursively enumerable languages is very broad and properly includes the CFL's.

Instantaneous Description (ID):

Instantaneous description (ID) of the Turing Machine M

can be denoted by $\alpha_1 q \alpha_2$

where $\alpha_1, \alpha_2 \rightarrow$ the string in Γ^*



$$\alpha_1 = a_1 a_2$$

$$\alpha_2 = a_3 a_4 a_5$$

Moves in a TM:

The change of ID is represented by:

$$x_1 x_2 \dots x_{i-1} q x_i \dots x_n \vdash x_1 \dots x_{i-2} p x_{i-1} x_i x_{i+1} \dots x_n$$

①. Turing machine with stay option:

TM with stay option can be defined as...

$(Q, \Sigma, \delta, q_0, F, B, \Gamma)$.

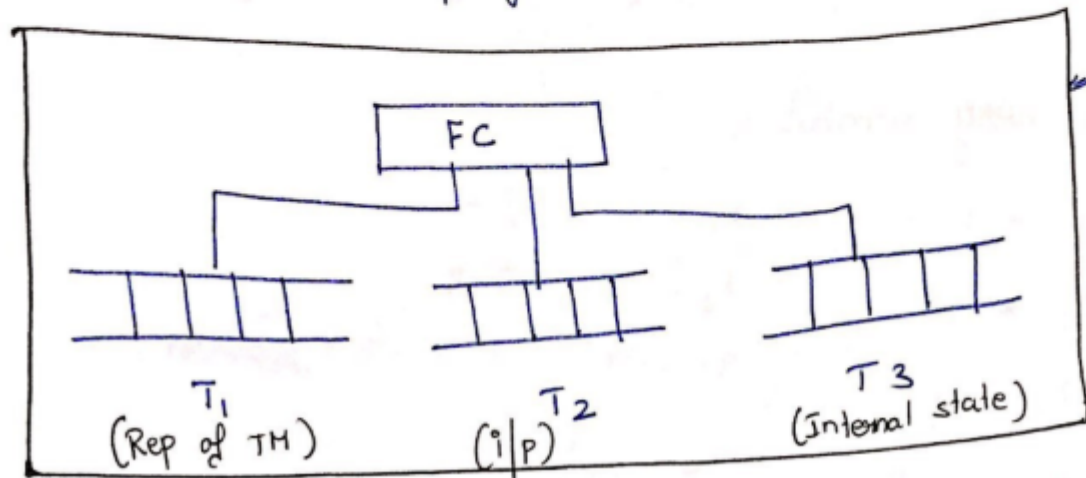
where $Q, \Sigma, q_0, F, B, \Gamma$ have same

meaning as std. TM.

But $\delta: Q \times \Gamma \rightarrow Q \times \Gamma \times \{L, R, S\}$

Universal Turing Machine:

- * It will have multitape TM.
- * It is reprogrammable.



Use this as computer then it should do any generic work

Tape T_1 in UTM contains the representation of

some TM
↓ (eq)

Tape T_2 in UTM contains the original i/p given to

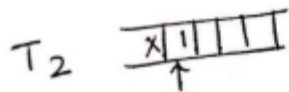
some TM
↓ (eq)

Tape T_3 in UTM contains the internal states (initially starts with start state)

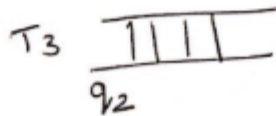
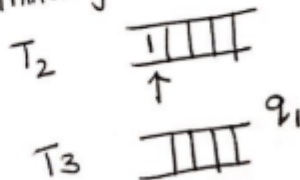
Ex:
$$\frac{S(q_1, 1) = (q_2, x, R)}{\Downarrow}$$

 T_1

after this transition



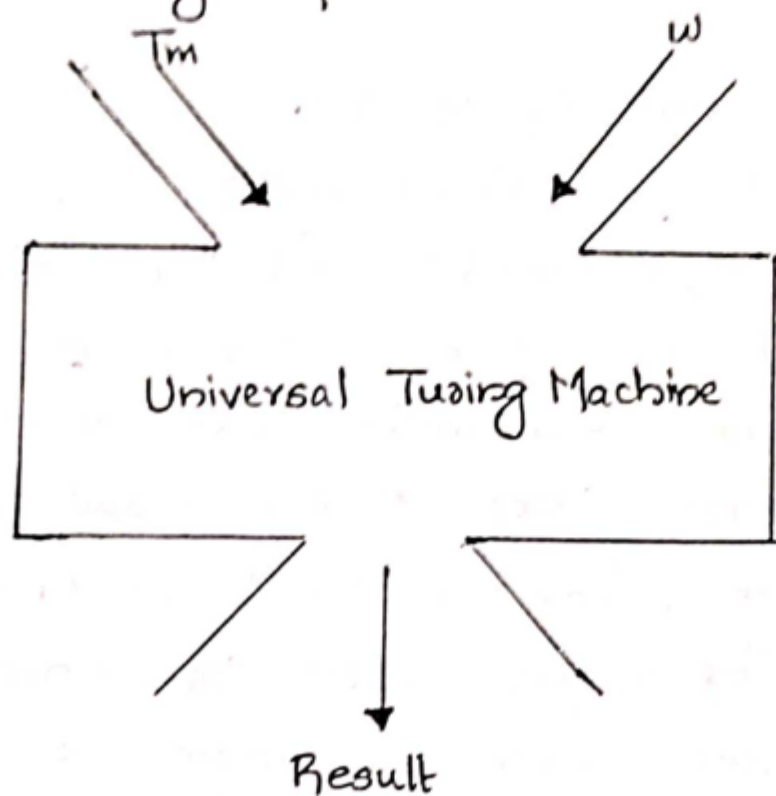
initially



Universal TM 'U' takes two arguments, a description of a machine T_m , " T_m " and a description of an input string w , " w ". We want U to have the following property:

U halts on input " T_m " " w " if and only if T_m halts on input w .

$U("m" "w") = "m(w)"$ is the functional notation of universal Turing Machine.



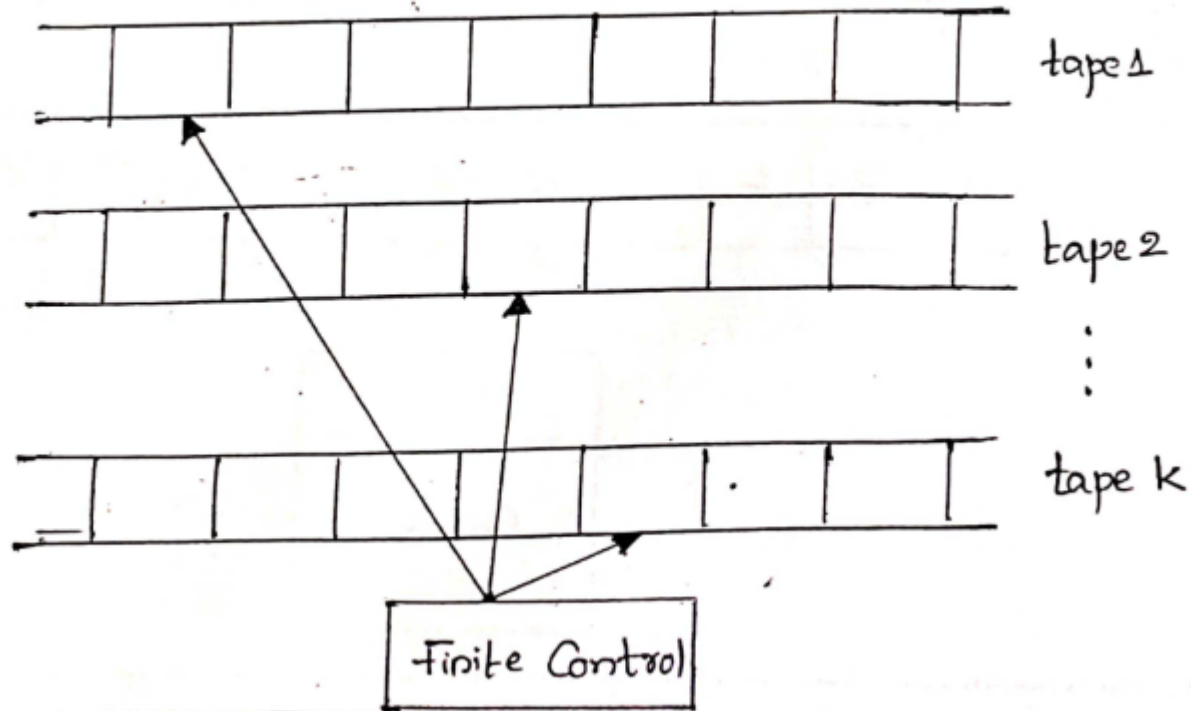
Universal Turing machine interprets all Turing machine

T_m : Desc of Turing machine

w : Input String

U halts for T_m, w if T_m has final state halt for w

(ii) Multi Tape TM

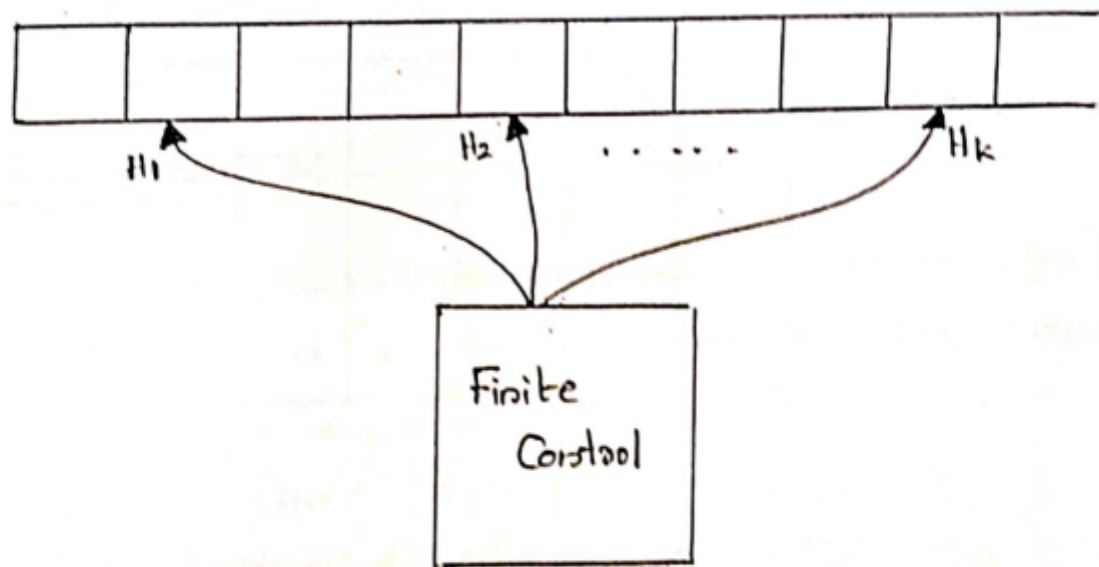


As shown in the figure a Multi-Tape TM consists of a finite control with k heads and k -tapes. Each tape can be either finite or infinite. On a single move depending on the state of finite control and the symbol scanned by each of the tape heads the machine can:

1. Change state.
2. Print a new symbol on each of the cells scanned by its tape heads.
3. Move each of its tape heads independently, one cell to the right or left or keep it stationary.

A k -Tape TM is formally defined as $M = (Q, \Sigma, \Gamma, \delta, q_0, B, F)$ where Q is the set of states, Σ is the set of i/p symbols, Γ is the set of tape symbols, δ is the transition function that maps from $Q \times \Gamma^k \rightarrow Q \times (\Gamma \times \{L, R, N\})^k$, q_0 is the start state, B is the blank symbol and F is the set of final states.

(iii) Multi Head TM.

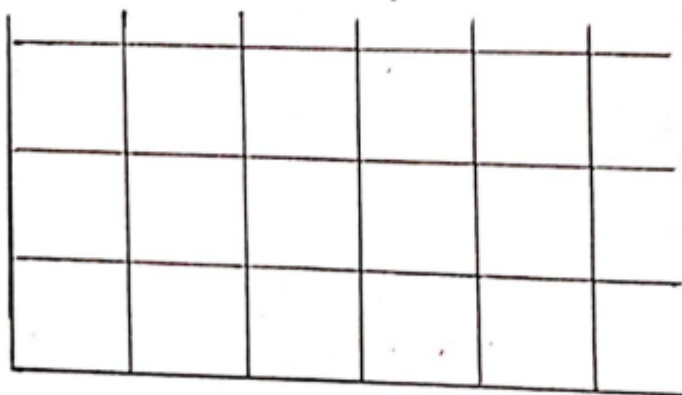


As shown in the figure a k head TM has a fixed no. of k heads. The heads are numbered 1 through k and a move of TM depends on the current state and on the k symbols scanned by each head. In one move, the heads may each move independently left, right or remain stationary.

A k -Head TM is formally defined as
 $M = (Q, \Sigma, \Gamma, \delta, q_0, B, F)$ where Q is the set of states,
 Σ is the set of i/p symbols, Γ is the set of tape symbols,
 δ is the transition function that maps from $Q \times \Gamma^k \rightarrow Q \times (\Gamma \cup \{ \square, \uparrow, \downarrow \})^k$,
 q_0 is the start state, B is the blank symbol and F is the
set of final states.

(iv) Multi Dimensional TM

Multi Dimensional TM has the usual finite control, but the tape consists of a k -dimensional array of cells, infinite in all $2k$ directions for some fixed k . Depending on the state and symbol scanned, the device changes state, prints a new symbol and moves its tape head in one of $2k$ directions, either positively or negatively along one of the k axes. Initially the i/p is along one axis and the head is at the left end of the i/p. At any instant, only a finite number of rows in any dimension contain non blank symbols and these rows each have only a finite number of nonblank symbols. The following figure represents a 2-D tape.



Let $q \in Q$ and $C_k \in \Gamma - \{B\}$

The configuration at a particular instant is denoted by $(q, (H_1, H_2), (i_1, j_1, C_{i_1 j_1}), (i_2, j_2, C_{i_2 j_2}), \dots, (i_k, j_k, C_{i_k j_k}), \dots)$

(ii) Offline TM

An Offline TM is a multitape TM whose i/p tape is read only. Usually we surround the i/p by endmarkers, $\$$ on the left and $\$$ on the right. The TM is not allowed to move the i/p tape head off the region between $\$$ and $\$$. It should be obvious that the offline TM is just a special case of multitape TM and therefore is no more powerful than any of the models we have considered.

(i) Non Deterministic TM.

A Non Deterministic TM is a device with a finite control and a single one way infinite tape. For a given state and tape symbol scanned by the read/write head the machine has a finite number of choices for the next move. Each choice consists of a new state, a tape symbol to write and a direction of head movement. The NDTM accepts its input if any sequence of choices of moves leads to an accepting state. Non determinism allows parallel computation.

* finite no. of choices for the next move

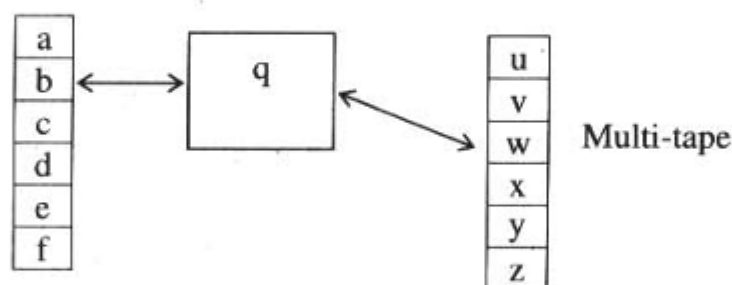
A NonDeterministic TM is formally defined as $M = (Q, \Sigma, \Gamma, \delta, q_0, B, F)$ where Q is the set of states, Σ is the set of i/p symbols, Γ is the set of tape symbols, δ is the transition function that maps from $Q \times \Gamma \rightarrow$ power set of $(Q \times \Gamma \times \{L, R, N\})$, q_0 is the start state, B is the blank symbol and F is set of final states.

7.9. Equivalence of Single Tape and Multi-tape TM's

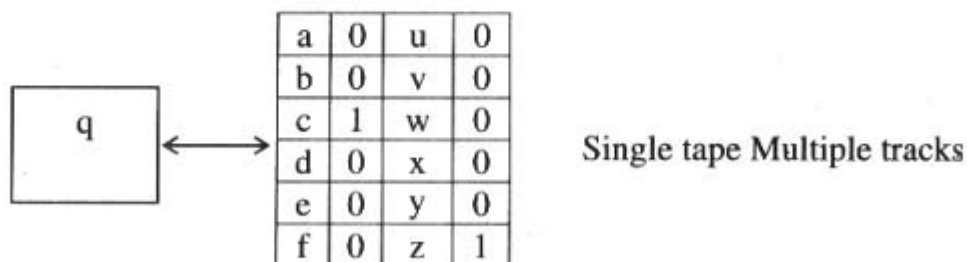
❖ **Theorem:** Every language accepted by a multi-tape TM is recursively enumerable.

Note: The theorem clearly indicates that every language accepted by a multi-tape TM is also accepted by a standard TM.

Proof: This can be shown by simulation. For example, consider a TM with two tapes as shown below:



The above 2-tape TM can be simulated using single tape TM which has four tracks as shown in figure below:



The first and third tracks consist of symbols from first and second tape respectively. The second and fourth track consists of the positions of the read/write head with respect to first and second tape respectively. The value 1 indicates the position of the read/write head. It is clear from the above figure that, the machine in state q and when the first read/write head points to the symbol c , the second read/write head points to the symbol z , then the machine enters into state p , if and only if this transition is defined for TM with multi-tapes. So, whatever transitions have been applied for multi-tape TM, if we apply the same transitions for the new machine that we have constructed, then the two machines are equivalent.